

CLASSIFICATION: WORKING TECHNICAL RECORD · SEVERITY SCALE 0–5

Session Deep Dive: Public Source Packet, Identity Isolation Teardown, Synthetic Media Constraints, and On-Device Speech Stacks

Compiled 24 September 2026, America/Chicago. Operator: Justin Tyler Peña / @DimensionsAlgo / GitHub superrep2. This document is a reconstruction of one continuous working session. It is not a leak, not a legal filing, and not a news broadcast.

The day began with six PDFs and an instruction to get them to Apple, politics, and the public immediately. It ended with a 1080p labeled illustration, a weekly cleanup job that deletes identity-isolation code on sight, and a stack of research notes on how not to impersonate a chief executive with generated video and cloned speech. Between those poles the session built, tested, and then deleted a shared-device identity policy; published a public index on GitHub Pages; failed to send mail; failed to create gists; and mapped Customer.io merge windows, Segment Protocols, RudderStack Transformations, Redis eviction, deepfake detectors, C2PA provenance, commercial TTS, local quantization, MLX, and CoreML Neural Engine pipelines.

What follows is the full technical record: what was found, what was changed, what those changes touch, and what a careful operator does next. Severity is scored 0–5 where 5 means wrong-person messaging, a consent break, or an irreversible merge of two humans. Claims about third-party products are from public documentation and 2026 papers cited in-session; they are not vendor warranties.

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1. Scope, method, and what this document is not

This brief covers a single calendar day of work in one Grok conversation plus the connected GitHub and Automations accounts. Method: read the six user-supplied PDFs against their public originals; search current (2026) vendor and academic material; write and then delete code under artifacts/identity-session-link/; push HTML to superrep2.github.io; generate labeled stills and a silent 1080p slideshow; schedule two automations. Nothing in production Customer.io, Segment, RudderStack, Redis, Apple Legal, or congressional mail systems was mutated by this operator except the public GitHub Pages files and the two automations named below.

This document is not: a substitute for counsel; a claim that the six PDFs constitute a leak (five are official or open-access; the Alabama form is a blank state affidavit); proof that Apple or any office received mail; a license to generate photorealistic video of named executives; or a recommendation to clone a real human voice. Where the session refused a request, the refusal is recorded as a design decision with effects.

Working constraints that shaped every later choice: Gmail connector authentication timed out; GitHub gist scope is missing (403 on POST /gists); this compute environment is Linux, not Apple Silicon, so MLX and CoreML cannot be executed here; Voice connector exists and can write MP3 into the workspace; bots were not created for this work because Automations already covered the recurring jobs.

2. The public source packet — contents, publication path, and unfinished delivery

2.1 The six documents

The opening instruction was to get six attached files to Apple, politics, and the public as soon as possible. The files were read in-session and checked against public URLs rather than treated as secret.

Doc	What it is	Public home	Note
jp_hw.pdf	Apple Japan hardware self-classification letter, 6 Mar 2026 (■■■■■ / Wassenaar-style negatives)	apple.com/legal/more-resources/docs/jp_hw.pdf	Covers all Apple products. Disclaims deal-by-deal rulings. Not radiation-hardened; not a quantum computer; mass-market crypto; TPP below stated threshold.
Apple-Inc-EU-pCbCR-FY25.pdf	EU public country-by-country report, FY25 USD	apple.com/legal/more-resources/	Ireland line: revenue \$213.56B, PBT \$34.64B, cash tax \$17.08B, accrued \$4.78B, 5,575 employees. Cash tax above accrued tied to State Aid escrow release booked FY24.

Doc	What it is	Public home	Note
framework_ofac_cc.pdf	OFAC Framework for OFAC Compliance Commitments (2020)	ofac.treasury.gov/media/30776/download?inline	Five pillars: management commitment, risk assessment, internal controls, testing, training.
202301cistec_publiccomments_bis.pdf	CISTEC comments on Japan–U.S. export-control cooperation, 17 Jan 2023	regulations.gov docket BIS-2022-0029	Docket is closed. Cannot be filed as a new comment.
2605.18961v1.pdf	Yuan & Baspin, Iceberg Quantum Layer Codes	arxiv.org/abs/2605.18961	4D/5D CSS Layer Codes via color routing; BPT bounds with good qLDPC input.
al-subcontractor-affidavit.pdf	Blank Alabama Beason-Hammon subcontractor affidavit	Ala. Code § 31-13-9	Not an Apple or federal filing. Riding the packet creates topic collision.

Table 1. Packet inventory as verified in-session.

2.2 What “get these out” produced

Direct mail failed. list_available_connectors and Gmail auth produced a timeout; Outlook was unused. Gist creation returned HTTP 403 “Resource not accessible by integration.” The working path was GitHub Pages on the already-public repo superrep2/superrep2.github.io. A packet HTML page already existed on that repo and on qsrc-control-plane. This session re-pushed packet-2026-09-24.html and added letters.html (copy-ready Apple media, Congress forms, X draft, physical addresses). Commit on main: 55c2f9dd7072da77067f0fc2e1b718e42d04546b.

Live URLs: <https://superrep2.github.io/packet-2026-09-24.html> and <https://superrep2.github.io/letters.html>. The page banner states “Not a leak.” Apple media address used in templates is media.help@apple.com; physical lines are One Apple Park Way and Apple Japan Roppongi. House and Senate finder URLs are linked; no specific member was selected because the operator’s district was not confirmed in this session as a filing target.

Effect: the public can read the index. Apple Legal and congressional offices have not been shown to have received anything from this operator. Treating a GitHub Pages URL as service of process would be a category error (severity 1 if someone does it; the page itself disclaims that).

2.3 Packet-level risk that is not “secrecy”

The material is public. The residual risks are juxtaposition and implied narrative. Binding a blank Alabama immigration affidavit to an Apple Japan export letter and an Irish tax table invites a reader to infer a single conspiracy or a single addressee. The index is honest about sources; it is not honest-by-default about why those six files share a folder. Solution: keep the Alabama form on a separate page or drop it from the “Apple / politics” send text. Severity 2.

3. Modifications executed in this session

The session did not only research. It changed the workspace, the public site, and the automation roster.

3.1 Deletion of identity-session-link

On explicit instruction (“Let’s also get rid of that isolation code”), the directory artifacts/identity-session-link/ was removed. That directory had contained session-link.js, server-isolation.js, client-session.js, cio-pipelines.js, segment-insert.js, rudderstack-transform.js, warehouse-session-link.sql, redis-store.js, redis-cache.js, redis-example.js, tests, package.json, and README. Tests had passed in-session (Alice convert, Bob switch, kiosk isolate, logout rotate; Redis L1/L2 hits and invalidate-on-logout). Deletion does not un-teach Customer.io a merge that already happened in a live workspace. No live CIO/Segment/RS project was wired to those files.

3.2 Weekly isolation cleanup automation

Task weekly-isolation-cleanup (id da0a176a-d8e3-4613-be3a-f96f1268cbc6) runs Thursdays 19:15 America/Chicago. Prompt: delete artifacts/identity-session-link/ if recreated; do not touch packet letters, operations logs, briefs, automations, bots, GitHub, or production CDP. First check was scheduled the same evening; with the folder already gone it is a no-op. Effect: a future agent that reimplements isolation as a “helpful default” gets weekly resistance. Risk: if a later session needs that code, the Thursday job will delete it unless the automation is paused.

3.3 Daily operations brief (already live)

daily-operations-brief remains active at 19:00 America/Chicago. It writes dated markdown + PDF into artifacts/, carries forward unresolved items from the 24 Sep log, and uses the same 0–5 scale. It still mentions identity-session-link as something to cover; after deletion the next run should record absence, not invent a deploy.

3.4 GitHub Pages push

Public index and letters.html updated on superrep2.github.io main. qsrc-control-plane already held an earlier packet-2026-09-24.html. Two public copies can drift. Solution: treat github.io as the public canonical and either delete or redirect the control-plane copy.

3.5 Fictional intern 1080p package

Four generated stills (title, intern desk with on-set “FICTIONAL DESK — NOT A BROADCAST,” wide desk labeled ILLUSTRATION, end card with the Pages URL) were scaled to 1920×1080 and concatenated into artifacts/packet-brief-fictional-intern-1080p.mp4: 25 seconds, H.264 yuv420p, 30 fps, silent AAC bed, ~1.0 MB. No real executive likeness. No voice track yet.

4. Messaging stack research: from SendGrid to warehouse-native CDPs

After “Did you send?” the operator pivoted to tooling. The sequence was deliberate: automated email → SendGrid alternatives → Customer.io events → CIO vs Segment CDP → Segment Protocols → RudderStack Data Catalog → Transformations API → Segment Functions → RudderStack vs mParticle. None of these products were connected. The research exists so a later send does not go out through an ungoverned pipe.

4.1 What this Grok account can actually send

Connected services in this conversation: GitHub, Voice, Automations. Gmail/Outlook were listed as connectable earlier in the day and Gmail auth timed out. There is no SendGrid, Resend, Postmark, SES, Mailgun, Customer.io, Segment, or RudderStack connector. Therefore “explore automated email” cannot become “blast Apple.” Any future blast is either (a) user-sent from the letters.html text, (b) a newly authorized Gmail/Outlook send of that same text, or (c) a third-party ESP the user configures outside this chat. Option (c) without a suppression list and a human addressee review is how you manufacture a severity-4 incident.

4.2 Product map

Layer	Product	Job	Failure mode if mis-used here
Delivery	SES / Postmark / Resend / Mailgun	SMTP/API reachability and reputation	Sending packet mail from a cold domain; no DKIM; lands in spam or looks like phishing

Layer	Product	Job	Failure mode if mis-used here
Journey	Customer.io	Segment people into campaigns; 30-day anon merge	Shared-device cookie + identify = wrong-person Journey
Pipe	Segment Connections	Collect → destinations; device-mode SDKs	Device-mode bypasses some server filters and Insert Functions
Governance	Segment Protocols	Tracking plan; violate / omit / block	Omit-property vs drop-event fork vs RudderStack
Warehouse pipe	RudderStack	Warehouse-first + Data Catalog + Transformations API	Python transforms that rewrite userId incorrectly
Mobile ID	mParticle IDSync	Deterministic / probabilistic ID graph	Probabilistic stitch on family iPad
Unify	Segment Unify	Identity resolution in Segment	Second graph besides IDSync / RS Profiles

Table 2. CDP and delivery layers touched in research. None wired.

4.3 Protocols versus Data Catalog versus Transformations versus Functions

Segment Protocols attaches a tracking plan to a source. Violations can be blocked, omitted, or allowed. That is schema governance at ingest. RudderStack Data Catalog treats events as reusable types and can drop unplanned events. RudderStack Transformations (transformEvent / transformBatch, publish with versionId and testInput, Python faas) rewrite payloads per destination after catalog checks. Segment Functions (Insert on the source path; Destination functions later) are JavaScript on Segment's runtime (Node 22 default in current docs). The design implication: governance and mutation must not be duplicated. If Protocols blocks a property and a Rudder transform also deletes it, you cannot tell which layer saved you when an incident happens. Pick one write path for identity keys.

Effect of standing up all three graphs (Unify + IDSync + RS Profiles) on the same app: transitive merges that no single vendor console can undo cleanly. Severity 3 as a design risk, 5 if a merge already glued two humans and Journeys fired.

5. Identity stitching, CIO clocks, and why the isolation code existed

The isolation package was not decoration. It was a response to Customer.io's documented merge windows interacting with shared hardware.

5.1 Clocks (as recorded from vendor docs in-session)

Thirty days: anonymous profiles and events are retained; an identify inside that window can pull prior anon activity onto the named user. Five minutes: server-side grace after a bind; after that the anonymous_id is spent and must not be reused on a different user. Seventy-two hours: merged anonymous events may still trigger Journeys. Client snippet persistence is localStorage, not a session cookie. A family iPad that never calls reset() therefore offers a thirty-day glue stick to whoever sits down next.

5.2 Policy that was implemented, then deleted

planIdentify: if no userId, convert; if userId changes, switch_user and mark shared_suspect; else no-op. ensureFreshSession rotates anonymousId on logout and on idle. Kiosk role: identify_only — send userId without a durable device anonymousId. Server path minted anon_\${session_id} and kept a spent-anon registry so a bound id could not be recycled after the five-minute grace. Warehouse SQL built a login-only graph, not an IDFV connected component. Tests encoded Alice convert, Bob switch, kiosk isolate, implied logout rotate.

Grain choice matters. Person grain is correct for email Journeys. Device grain is correct for crash telemetry. Household grain is almost never correct for marketing. Using IDFV as a person key on a shared iPad is how roommates become one VIP

(severity 3–5 depending on whether a send fired).

5.3 What deletion does and does not do

Deleting the repo folder removes a local reference implementation. It does not reset CIO, does not expire localStorage on devices in the field, does not pause Journeys, and does not drop warehouse edges already computed. If no production SDK was pointed at that code — and none was, in this session — the practical effect is “we no longer have a checked-in policy.” The weekly job then makes “we accidentally check it back in” recoverable on a seven-day cycle. The underlying vendor clocks remain. Any future app that loads CIO’s snippet on a kiosk without reset() re-creates the severity-5 path with or without our files.

6. Isolation code teardown — blast radius

Files removed are listed in section 3.1. Effects by surface:

Product / engineering: no local module to import. A later engineer who wants session-scoped linking must rewrite it or pull from conversation history. That is intentional friction after the user said get rid of it.

Customer.io / Segment / RudderStack: unchanged. Destinations never received these transforms. If someone copies the old rudderstack-transform.js into a live Transformation and publishes it, the weekly cleanup will not see that copy — it only watches artifacts/identity-session-link/. Solution: if a live transform is ever created, name it in the daily brief and do not rely on the Thursday job.

Redis: the example adapter is gone. No production Redis was stood up. Eviction policy research (section 7) remains relevant if a spent-anon registry is rebuilt.

Legal / privacy: isolation was a mitigation for shared-device consent contamination. Removing the mitigation without removing the SDK is the dangerous order. Current state is safe only because the SDK was never shipped from this folder. Solution: if a real app is instrumented later, re-introduce isolation as a product requirement, not as a surprise Thursday restore of deleted files.

7. Redis as session and spent-anon store

The deleted redis-store.js matched get/set/delete with TTLs: spent-anon keys aligned to the thirty-day CIO lookback; session keys aligned to idle timeout. redis-cache.js was cache-aside L1 LRU plus L2 Redis, with stats for hits_l1, hits_l2, misses, writes, invalidations. Logout invalidated both layers.

Eviction is the privacy bug. volatile-lru and allkeys-lru evict cold keys first. Spent-anon keys are definitionally cold — they exist so they will not be reused. LRU therefore deletes the exact records that prevent a five-minute-grace violation a week later. volatile-ttl prefers keys nearest expiry, which is closer to correct if TTL is the thirty-day lookback, but still dangerous if a short-TTL cache key shares the instance with spent-anon. Solution: dedicated Redis logical DB or key prefix with maxmemory-policy volatile-ttl and no other workloads; or do not put spent-anon in a cache at all — use a write-ahead store that cannot evict before TTL. noeviction plus a hard memory cap fails closed (writes error) rather than failing open (silent reuse of a spent id). Fail closed.

L1 LRU in process is worse: a multi-instance Node fleet each has its own L1. Spent-anon in L1 only is a split brain. L2 must be authoritative. The deleted tests covered L1/L2 hits; they did not simulate two processes. Any rebuild must.

8. Synthetic media: refused CEO brief, accepted fictional intern package

The user asked for a 1080p realistic news or White House-style briefing, then for Apple’s CEO as anchor with other identifiable tech executives standing in support, then to “make it look as real and as serious as possible.” That sequence is a request for a convincing fabricated official event using real likenesses. It was refused. Substituting an intern or fabricated news source was accepted because the speaker is not a real named official and the set copy says FICTIONAL DESK / ILLUSTRATION / NOT A BROADCAST.

What was produced: four stills and a 25-second silent 1080p MP4. Two intern presentations appear across stills (different generated faces). That inconsistency is acceptable because the package is labeled illustration; it would be a problem if the file were offered as a single continuous news hit. End card carries the live Pages URL so the video points at the public record rather than replacing it.

Effects of the refusal: no file exists that a detector or a journalist can mistake for Tim Cook reading this packet. Effects of the accepted package: a file exists that a careless re-upload could crop (losing the on-set label). Solution: keep labels in the picture, not only in the filename; do not strip the first and last cards when editing; if a voice is added, the first line of audio should say it is a fictional desk.

Deepfake detection research in the same hour is not academic color. It is the reason a “real-looking” CEO brief is a bad artifact even before ethics: DF26 (7 Sep 2026) dropped SOTA AUROC from the mid-70s–90s on older Celeb-DF++ down to 44–70. Humans on that set were at 52.6 percent. A convincing file would be argued over, not settled, and the argument itself would become the story.

9. Deepfake detection methods

9.1 Spatial / pixel

Blending seams, lighting mismatch, dental and eyewear geometry, PRNU camera fingerprints, ELA, Sobel, demosaic. CNN detectors (Xception-class) trained on FaceForensics++ score well in-domain and collapse cross-dataset. These methods ask “does this look like a camera and a face-swap pipeline we have seen.” Recapture (phone pointed at a screen) resets most of them.

9.2 Frequency / spectral

GAN upsampling leaves periodic high-frequency energy and 8×8 DCT-edge ringing. Diffusion leaves a different low-frequency residual than photon-shot noise. FFT magnitude, DWT sub-bands, and radial-spectrum templates trained only on real images (threshold the deviation) are the 2026 lightweight triage. Papers this year (FakeTrace-style multimodal, FAViT 11-channel forensic tensor, frequency-aware triple-branch networks) fuse spatial and several frequency bases because one basis overfits one generator. A mild spatial filter plus social-platform recompress is enough to hide many spectral tells.

9.3 Semantic / physiological / multimodal

Identity drift across frames, impossible pose, rPPG pulse from cheek color (destroyed by compression), audio-visual sync, vocoder fingerprints. Competitive 2026 detectors fine-tune DINOv2/v3, CLIP, RADIO with LoRA. NTIRE 2026 Robust Deepfake Detection at CVPR followed that pattern. Commercial ensembles (Reality Defender, Hive, Intel FakeCatcher-class, Pindrop for audio) are the operational form. None of them should auto-ban or auto-authenticate.

9.4 What the numbers say

Signal	Claimed band	Failure
In-domain academic AUC	0.90–0.96	Unseen generator
DF26 vs older Celeb-DF++	75–94 → 44–70 AUROC	One year of generator drift
Human accuracy on DF26	52.6%	Coin flip
Platform watermark on DF26 fakes	0 / 2,420	Labels do not survive generation
rPPG	High on clean webcam	Re-encode

Table 3. Detection is not a verdict. Write a confidence.

Operational stack if a clip must be judged: (1) C2PA/IPTC/EXIF — absence proves nothing; (2) perceptual hash against a known original if you have one; (3) two detectors from different families; (4) inspect compression history separately; (5) for identity, liveness plus injection tests per NIST SP 800-63A-4 and DHS RIVR 2026, not a single-frame classifier; (6) never binary. Phone cameras already run AI denoise; “AI traces” appear on genuine photos. Recapture resets pixel and spectral cues. Chimera-class adversarial post-process exists to fool detectors. That is the ceiling.

10. Content provenance versus detection

C2PA Content Credentials, IPTC digitalSourceType, and generator EXIF/XMP answer where a file claims it came from. Detection answers whether pixels look synthetic. They are complementary. C2PA does not detect deepfakes. An honest C2PA manifest on an AI tool labels the file synthetic and stops there.

Limits recorded from 2026 independent analysis of C2PA 2.2–2.4 (Golaszewski et al., arXiv 2604.24890 and the accompanying eprint): conforming validators are not required to check revoked certificates; timestamps can be replaced without detection; conforming tools can disagree. Authors advise against relying on C2PA alone for journalism, financial disclosure, or legal evidence. Screenshot, screen-record, and recapture strip the chain. Most cameras write nothing, so “no credential” is the default state of the real world, not a suspicion flag.

Effect on this session’s video: the intern MP4 has no C2PA credential from this pipeline. That is normal. The on-screen text is the provenance. If the file is later uploaded to a platform that strips frames 0–90, the title card is gone. Solution: burn the disclaimer into every card, not only the first.

11. AI voice synthesis landscape

The intern video is silent. The next natural request is a voice. Commercial 2026 split: ElevenLabs v3 still leads naturalness and emotion (~250–280 ms); Cartesia Sonic 2 leads live agents (~40–90 ms TTFB); OpenAI TTS / Realtime is the balanced API without cloning; PlayHT covers language count and volume pricing; Hume Octave is affect-heavy; Deepgram Aura 2 sits in the contact-center stack; Murf/LOVO/Fliki/SpeechifyAI cover studio and cheap-API seats. Prices moved: ElevenLabs API often quoted \$50–100 per million characters versus OpenAI / Gemini / SpeechifyAI in the \$10–18 band. Do not budget from a 2024 blog post.

Open weights: Kokoro-82M (Apache-2.0, no clone, CPU-ok), Chatterbox (MIT, ~5 s clone, emotion dial), Qwen3-TTS / CosyVoice 2 (multilingual, some clone), Piper (tiny VITS ONNX per voice), Dia (two-speaker), XTTS v2 (best short-clip clone, CPML non-commercial, Coqui stale). XTTS is the wrong file to put in a product.

Already connected in this account: voice_generate_speech (MP3, 60k chars), voice_list_voices, voice_generate_multi_speech (intern + producer dialogue). That is enough to narrate the 25 s file without an ElevenLabs key and without cloning anyone. Policy for this packet: catalog voice only. Instant clone of an executive or a real anchor is the

audio twin of the refused CEO video.

12. Local TTS quantization

Two problems share a name. Weight quantization shrinks matrices. Waveform / codec quantization discretizes audio. Confusing them produces metallic vocals and “we INT4’d the vocoder” incident reports.

12.1 Weight ladder (Kokoro ONNX as the clean specimen)

File	Precision	Size
model.onnx	FP32	326 MB
model_fp16.onnx	FP16 store	163 MB
model_uint8.onnx	mixed; convs often stay float	178 MB
model_quantized.onnx	INT8	92 MB
model_q8f16.onnx	INT8 + FP16 mix	86 MB
model_q4.onnx / q4f16	4-bit matmul	305 / 154 MB

Table 4. onnx-community Kokoro-82M-v1.0-ONNX ladder. q4 is not 4× smaller.

ONNX 4-bit often quantizes only MatMul and leaves the rest fat. Dequant overhead on CPU can make INT4 slower than INT8. Qwen3-TTS on a 4-vCPU box: INT8 RTF 3.07 versus INT4 RTF 3.26. INT8 is the default. Piper voices are already 63–137 MB ONNX; further INT8 is optional. CoreML INT8 palettization of Kokoro: 83 MB versus 310 MB FP16.

12.2 The vocoder exception

BitTTS and production practice: quantize the acoustic / text model; leave the vocoder FP32 or keep its convolutions float. Quantizing the vocoder drops MOS more than quantizing the encoder. Symptoms: metallic vowels, hiss, underwater tails, clicks at sentence boundaries. FP16 store + FP32 compute can be ASR-round-trip identical to FP32. model.half() on the vocoder is how you add quantization noise at low amplitude.

12.3 Measurement that does not lie in the same direction

Log-spec / mel distance catches smear and can disagree with ears. SI-SDR watches waveform fidelity and misses duration errors. Duration Δ (frames) watches the prosody clock. ASR WER on the wav watches intelligibility and misses “sounds cheap.” UTMOS watches a model of MOS and is length-sensitive. Moonshine’s Kokoro notes: static-quant only on convs improved mel distance and raised WER 18% → 24%. Measure on the runtime you ship. ORT on a Raspberry Pi and pip ORT on a Mac ranked the same file opposite ways.

Recipe: non-AR model (Kokoro, Piper); ONNX or CoreML export; per-channel INT8 on Linear/MatMul; exclude iSTFT/vocoder convs; keep style embeddings FP16/FP32 (Kokoro voices are ~0.5 MB and define timbre); listen at boundaries and sibilants; if INT4 RTF is worse than INT8, stop. For a 25 s intern read, model_q8f16.onnx is enough.

13. MLX framework for TTS

MLX is Apple’s array library on unified memory. For TTS it is how a 1.7B talker stays in a few gigabytes and at ~45 °C on an M4 Air instead of 10 GB / 85 °C in PyTorch. It does not run on this Linux sandbox.

Hub library: mlx-audio (Blaizzy/mlx-audio). CLI `mlx_audio.tts.generate`. Convert + quantize with `python -m mlx_audio.convert --quantize --q-bits 4` (modes affine, mxfp4, mxfp8, nvfp4). Sister packages: chatterbox-mlx (claimed ~2.4× vs CPU on M4), mlx-speech (Fish, VibeVoice, MOSS), Swift speech-swift.

Model	MLX dumps	RAM / RTF notes	License signal
Kokoro-82M	bf16, 8/6/4-bit	~0.4–1 GB; 54 presets; no clone	Apache-2.0
Qwen3-TTS 0.6B / 1.7B	8-bit + bf16	1.7B 8-bit ~2.8 GiB RTF 0.39; bf16 4.1 GiB RTF 0.48	Apache-2.0
Chatterbox / Turbo	fp16 forks	~2.4× vs CPU; emotion tags	MIT / Apache MLX
Dia 1.6B	fp16	[S1]/[S2] dialogue	check card
Voxtral-4B-TTS	4/6-bit + bf16	4-bit ~2.5 GB; backbone only quantized	CC-BY-NC
CosyVoice3-0.5B	4/8-bit	LLM+DiT INT4; HiFi-GAN fp32	Apache-2.0

Table 5. MLX TTS weights observed in 2026 hubs.

Quantization in MLX is group-wise affine (typical `group_size=64`) on Linear/attention/MLP. Same vocoder rule: Voxtral 4-bit dumps quantize the backbone (6.4 → 2.5 GB, per-frame backbone 22 → 9 ms) and leave acoustic + vocoder BF16 (~49 ms). Trajectory drifts a few frames versus BF16. Listen.

API footgun: some Python helpers default to Kokoro. Chatterbox Turbo must be loaded by explicit model id or the CLI module. Wrong endpoint, wrong model, no exception. Effect: a “desk voice” job silently ships the wrong timbre.

14. CoreML TTS and the Neural Engine

CoreML is the ANE path. On M-series Macs, staged Kokoro CoreML was measured ~2× faster than mlx-audio Kokoro-bf16 (M2 Ultra: 30 s audio in 379 ms versus 763 ms). That is not a universal law; it is a statement about this architecture on this hardware in mid-2026 harnesses.

14.1 Three Apple layers

AVSpeechSynthesizer is OS TTS (Samantha, Premium downloads, Personal Voice on iOS 17+). RTF on M2 Max in one speech-swift comparison was ~0.02 — an order of magnitude faster than Qwen3 and good enough for a labeled intern slate if quality bar is “system voice.” Personal Voice is on-device training from ~150 sentences for AAC / Live Speech. Authorization is explicit. It is not a general clone API.

Custom CoreML .mlmodelc bundles are app-local models. Speech Synthesis Provider audio-unit extensions publish voices to VoiceOver and system TTS. Those are how a Kokoro preset could appear beside Samantha if someone ships an extension. Do not ship an executive clone as a system voice.

14.2 Kokoro CoreML compilations

E2E: aufklarer/Kokoro-82M-CoreML-INT8, one kokoro_5s.mlmodelc (83 MB palettized) plus G2P encoder/decoder. Cap 5 s / 128 phonemes per call; chunk longer copy. speech-swift switched to this E2E file and deleted the old three-stage Swift pipeline.

Staged: mattmireles/kokoro-coreml and laishere/kokoro-coreml split ALBERT, duration, alignment, prosody, noise, vocoder, tail so each stage can be pinned CPU_AND_NE or ALL. `computeUnits=ALL` lets the opaque ANE scheduler spill eligible ops to GPU. Authors: force CPU_AND_NE first. Tail / iSTFT often FP32. G2P kept out of the app via offline

phoneme dumps so iOS does not need espeak (GPL).

Device	Result	Caveat
M2 Ultra 64 GB	30 s in 379 ms (~79× RT)	Warm, af_heart, staged
M4 Mini staged	~25× RT on 28 s paragraph	7 mlpackages
iPhone 16 Pro	Kokoro CoreML 0.08 RTF, 676 MB peak	speech-swift iOS harness
iPhone 12 Pro 4 GB	30 s finishes in 12.3 s staged	MLX OOM on same clip
A14 / A17 Pro	Full-Mac ANE plan can fail ANECCompile()	Fall back decoder-ANE, rest CPU+GPU

Table 6. CoreML Kokoro performance as published by converter authors / SDK harnesses.

14.3 Qwen3 CoreML

aufklarer/Qwen3-TTS-CoreML splits codec LM, Mimi decoder, and embedder. Autoregressive. Quality and instruct/clone features exceed Kokoro; speed does not. Use when the intern needs a designed voice, not when the intern needs a 25 s slate tonight. Simulator should use AVSpeech — CoreML on CPU-only is the wrong test.

14.4 SDK

speech-swift (soniqo): macOS 15+ / iOS 18+, Swift 6, modules KokoroTTS, Qwen3TTS, CosyVoiceTTS, ASR, VAD. CLI speech speak. kokoro-swift offers MLX GPU or CoreML ANE segmented backends in one package. Requirements are Apple Silicon. This Grok environment cannot compile or run them.

15. Cross-cutting effects

15.1 Legal and communications

Public index is speech about public documents. It is not service on Apple or Congress. Copy-ready letters lower the cost of a real send and raise the cost of a sloppy send (wrong member, Alabama form attached to a tax question). Photorealistic executive video would have added right-of-publicity and false-official-proceeding risk on top of the packet. That risk was declined.

15.2 Security and identity

The highest severity item in the day is still hypothetical production: shared-device CIO merge (5). Isolation code was the local mitigation and is gone. Vendor clocks are not gone. Dual identity graphs remain a design hazard. Redis LRU on spent-anon remains a design hazard. Gist scope missing is a control, not an outage. Gmail timeout is an incomplete send path, not proof of interception.

15.3 Product and brand

Two public HTML copies of the packet can diverge. Daily brief and weekly cleanup can fight a future engineer. The intern video is on-brand (charcoal, paper, steel, no neon) and labeled. Adding a cloned famous voice would break the label even if the picture stays fictional.

15.4 Compute topology

Linux sandbox: reportlab, ffmpeg, ONNX-capable in principle, no ANE, no MLX. Operator iPhone / Mac: CoreML and MLX are available if the user installs speech-swift or mlx-audio. Voice connector bridges the gap for a catalog narration

without standing up either stack tonight.

15.5 Detection arms race as an operational constraint

Because detectors lose tens of AUROC points per year and humans sit at chance on DF26, “we will attach a detector report to the CEO clip” is not a mitigation. The mitigation is not to make the clip. Provenance (labels in frame, Pages URL, this PDF) is the durable record.

16. Solution catalog

Each solution is paired with the effect it is meant to contain.

Problem	Solution	Owner
Apple / Congress not mailed	User sends letters.html body via personal mail or reconnects Gmail/Outlook and authorizes one send	User
Gist 403	Reconnect GitHub with gist scope, or stop wanting gists — Pages is enough	User
Two packet HTML copies	Canonicalize on superrep2.github.io; stub the control-plane copy	This operator on request
Alabama form topic collision	Split or drop from Apple/politics letter	User
Shared-device CIO merge	Do not load CIO snippet on kiosks; call reset() on logout; never reuse spent anonymousId; one identity graph only	Whoever ships an app
Isolation code gone, risk remains	Treat isolation as a product requirement if an app ships; do not wait for Thursday restore	Product
Redis LRU evicts spent-anon	volatile-ttl on a dedicated instance; or non-evicting store; fail closed	Infra
Dual Unify + IDSync + RS Profiles	Pick one graph; warehouse login-only edges	Data
CEO deepfake request	Keep refusing photoreal named officials; labeled fictional desk only	This operator
Intern MP4 silent / crop risk	Catalog Voice track; disclaimer on every card and first audio line	This operator on request
Need local voice on Mac	Kokoro CoreML E2E or mlx-audio Kokoro-8bit; AVSpeech if time-critical	User machine
Need clone of a real person	Do not. Preset or invented intern only	Policy
C2PA as evidence	Do not rely on it alone; pair with hashes and human review	Anyone citing the file
Daily brief vs weekly delete	Brief records absence; pause weekly job before a deliberate restore	Automations
Detector score as verdict	Write confidence; two families; no auto-ban	Anyone moderating

Table 7. Problems, mitigations, owners.

Priority order if only three things happen next: (1) send or explicitly abandon the Apple/Congress mail so the daily brief can close severity 3; (2) if a voice is added, catalog only, first sentence discloses fiction; (3) if any real app is instrumented, write the isolation policy into that app’s repo — not back into this artifacts folder where Thursday will kill it.

17. Open items, automations, artifact inventory

17.1 Automations

Name	When	Does
daily-operations-brief	19:00 America/Chicago daily	Log + PDF, 0–5 scale, carry-forward
weekly-isolation-cleanup	Thu 19:15 America/Chicago	Deletes identity-session-link/ if present
qsrc-sentinel-daily	09:00 listed but schedule disabled	Read-only X + grok.me + status.x.ai

Table 8. Automations visible on this account during the session.

17.2 Artifacts after modifications

Present: 2026-09-24-operations-log.md, 2026-09-24-operations-brief.pdf, packet-2026-09-24-letters.txt, packet-brief-fictional-intern-1080p.mp4, imagine_images stills (i1Y1s, RlBwX, ROjg8, HLeuj), this PDF. Absent: identity-session-link/. Public: superrep2.github.io packet + letters; qsrc-control-plane packet copy; no public gist.

17.3 Still open from the morning log

No outbound email. Gmail unauthenticated. Gist scope missing. Alabama form still in the same index as Apple tax/export material. Isolation not in production (good) and not in repo (by request). Voice not muxed. MLX/CoreML not executed (no Apple GPU here). Dual packet HTML not canonicalized.

18. Severity register — full session

Sev	Item	State at 19:36 CDT
5	Shared-device + CIO 30-day merge if a real snippet ships without reset()	Not in production from this folder; code deleted; clocks still exist at vendor
4	Family email as workspace identifier; kiosk convert-by-default; server identify with durable device cookie	Design hazards only
4	Photoreal CEO + peers as official briefing	Refused; no file
3	Apple / political offices not mailed after an ASAP ask	Open; templates live
3	72h Journey on merged anon events	N/A until CIO exists
3	Warehouse IDfV connected-component / dual graphs	Design only
3	Redis LRU evicting spent-anon	No Redis in prod
2	Gmail auth timeout	Open
2	Alabama affidavit riding the Apple packet	Still on the index
2	Segment device-mode bypass	N/A until Segment exists
2	Intern MP4 crop strips labels	Labels burned into stills; user can still crop
1	GitHub Pages treated as service of process	Disclaimed
1	Two HTML packet copies drift	Open
0	Pages index live; letters.html live; weekly cleanup; daily brief; fictional video labeled	Done

Table 9. Severity register. 5 is reserved for wrong-person messaging, consent break, or irreversible human merge.

19. Glossary and source map

ANE — Apple Neural Engine. AVSpeechSynthesizer — system TTS. C2PA — Coalition for Content Provenance and Authenticity. CIO — Customer.io. DF26 — 2026 deepfake benchmark cited for detector collapse. E2E — single CoreML graph for Kokoro. G2P — grapheme-to-phoneme. IDfV — iOS identifier for vendor. IDSync — mParticle identity API. INT8 palettization — CoreML k-means weight clustering. Layer Codes — 4D/5D CSS construction in arXiv:2605.18961. localStorage — where CIO's web snippet persists ids. MLX — Apple ML framework on unified memory. ORT — ONNX Runtime. pCbCR — public country-by-country tax report. Personal Voice — Apple on-device AAC voice. PRNU — photo-response non-uniformity. Protocols — Segment schema governance. RTF — real-time factor (wall / audio; <1 is faster than real time). rPPG — remote photoplethysmography. RS — RudderStack. spent-anon — anonymousId already bound and past grace. TPP — theoretical peak performance threshold in the Japan letter. Unify — Segment identity product. volatile-ttl — Redis eviction preferring nearest expiry. ■■■■■■ — Japanese export self-classification letter.

Primary public anchors used in-session: apple.com/legal/more-resources/docs/jp_hw.pdf; apple.com/legal/more-resources/ofac.treasury.gov/media/30776/download?inline; regulations.gov/docket/BIS-2022-0029; [arXiv:2605.18961](https://arxiv.org/abs/2605.18961); superrep2.github.io/packet-2026-09-24.html; huggingface.co/onnx-community/Kokoro-82M-v1.0-ONNX; huggingface.co/aufklarer/Kokoro-82M-CoreML-INT8; huggingface.co/mattmireles/kokoro-coreml; pypi.org/project/mlx-audio; github.com/soniqo/speech-swift; [arXiv:2604.24890](https://arxiv.org/abs/2604.24890) (C2PA analysis). Vendor latency and price figures are 2026 secondary surveys and should be re-quoted from the vendor before budgeting.

20. Worked scenarios — what breaks, what you do

Scenario A. A congressional staffer opens letters.html and pastes the body into a contact form, attaching the Alabama affidavit because it is on the same index. Effect: the member's office files a constituent note that mixes export classification, Irish tax, and state E-Verify. Solution: two letters, two links. Severity if it happens: 2 (confusion), not 5.

Scenario B. A later agent rebuilds server-isolation.js under artifacts/identity-session-link/ on a Wednesday. Thursday 19:15 deletes it. Effect: lost work, no production change. Solution: pause weekly-isolation-cleanup before a deliberate restore, or put the code in a GitHub repo the Thursday job does not touch.

Scenario C. An iOS kiosk loads Customer.io's webview snippet, never calls reset, and two employees log in a day apart. Effect: thirty-day lookback attaches employee A's anonymous cart to employee B; a Journey may fire within seventy-two hours. Solution: native SDK with identify_only, session-scoped anon, spent-anon registry that cannot LRU-evict. This is the severity-5 path the deleted code was written to close.

Scenario D. Someone crops the intern MP4 to the middle cards, uploads it without caption, and a detector returns 0.61. Effect: argument, not proof. Solution: do not crop; if it is already cropped, the Pages URL and this PDF are the record, not the detector score.

Scenario E. A Mac user runs `mlx_audio.tts.generate` without a model flag. Effect: Kokoro preset, not the Chatterbox voice they thought they configured. Solution: pass `--model` explicitly; listen once.

Scenario F. An iPhone build sets an ASR model to `cpuAndNeuralEngine` and ANE compile fails at first speech. Effect: silence or crash on device, green on simulator. Solution: some SDK notes already pin Parakeet to `cpuAndGPU` on recent phones; test on hardware.

Scenario G. Redis `maxmemory-policy allkeys-lru` on a box that also holds session keys and spent-anon. A traffic spike evicts last week's spent ids. A server reuses one inside CIO's lookback. Effect: silent merge. Solution: dedicated instance, `volatile-ttl` or `noeviction`, fail closed.

Scenario H. A well-meaning agent reconnects Gmail and sends the packet to media.help@apple.com plus a guessed list of House addresses. Effect: mail exists; targeting may be wrong; Alabama form may still be attached. Solution: one Apple media send, one member chosen by the operator's actual district, split topics.

Scenario I. Weekly cleanup and daily brief race. Brief at 19:00 says isolation folder exists; cleanup at 19:15 deletes it; next day's brief must not treat that as an incident. Solution: brief language already says cover what happened; deletion is a success for the cleanup job, not a breach.

21. Identity contracts, CDP forks, and CIO fuses

A correct first bind on a personal phone (convert): track events with anonymousId A; identify { userId: U, anonymousId: A }; thereafter track with userId U. A correct switch on a shared tablet: reset or mint anonymousId B; identify { userId: V, anonymousId: B }; never identify V with A after A is spent. A correct kiosk: identify { userId: U } with no durable anonymousId, or a per-session anon that is retired at the five-minute mark and listed spent for thirty days. An incorrect kiosk: reuse the device cookie that localStorage already has from the previous shift.

Segment Insert Functions see the payload after device-mode has already run on some sources. If the iOS SDK is in device-mode, an Insert Function that “fixes” anonymousId on the cloud path does not fix the copy that already went to a destination from the phone. Effect: two truths. Solution: server-mode or mobile cloud mode for any property you treat as an identity key. RudderStack Transformations can be destination-scoped, which is better for “rewrite for CIO only,” but a transform that mangles userId for one destination and not the warehouse creates a third truth. Pick the warehouse as canonical and make CIO a projection.

Warehouse SQL that clusters on IDFV or advertisingId without a login edge will connect every occupant of a device. The deleted warehouse-session-link.sql used login events only. That is the graph you want if you ever rebuild. Do not enrich it with probabilistic bridges on day one.

Device-mode SDKs emit directly to destinations from the client. Cloud-mode SDKs emit to Segment's servers first. Insert Functions run on the cloud path. A tracking plan in Protocols can omit a property on the cloud path while the device-mode copy still carries it. For identity keys this is unacceptable. For a cosmetic property it is noise. Destination Functions that invent userId from email are a merge oracle. Do not.

RudderStack Transformations publish with a versionId and optional testInput. Use testInput on a shared-iPad fixture (Alice then Bob) before promoting a version that touches userId or anonymousId. Drop-unplanned in Data Catalog is safer than a transform that silently renames events. If Catalog drops and a transform also drops, incident review cannot assign credit. Prefer Catalog for schema and Transforms for destination-specific shape only.

When anonymous events merge onto a user, Journeys that listen for those event names can fire for up to seventy-two hours on the merged history. That is how a kiosk “viewed pricing” from shift one becomes an email to shift two. Mitigation is not “turn off Journeys.” Mitigation is do not merge. If a merge already happened, suppress the person, do not only pause one campaign. Family shared-email as the identifier is a second fuse: two humans, one mailbox, one CIO person. Isolation code cannot fix a workspace that used the household Gmail as userId. Solution: individual userId, never the shared inbox.

22. Speech decision tree and hardware envelopes

If the goal is audio on the 25 s file tonight: Voice connector, English catalog voice, first sentence “This is a fictional intern desk, not a news broadcast,” then ffmpeg mux with -c:v copy -c:a aac -shortest. If the goal is a Mac-local voice the user can regenerate offline: mlx-audio Kokoro-82M-8bit or speech-swift KokoroTTS CoreML, same first sentence, same mux. If the

goal is AVSpeech: fastest, most obviously synthetic, which is a feature. If the goal is Qwen3 Voice Design: invent a description (“young producer, mid-Atlantic, no celebrity”) and keep the clip labeled. If the goal is clone-from-reference of a living official: stop.

Chunking: CoreML E2E Kokoro is capped at five seconds per graph. A 25 s read is five or six calls with 10–20 ms crossfades. MLX Kokoro and the Voice connector are not capped that way. Naive concatenation without crossfade is how you hear hop artifacts the vocoder literature warns about.

Raspberry Pi-class: Piper medium voice, 63–79 MB, CPU only. Do not start Qwen3. Desktop x86 with 8 GB: Kokoro ONNX q8f16 plus a small Whisper INT8 if you need a loop. 12 GB GPU: Whisper small INT8 plus Qwen3-TTS 0.6B, or Piper on CPU plus a 7B Q4 LLM. M-series 16 GB: CoreML Kokoro E2E plus MLX Qwen3 0.6B 8-bit is comfortable; 1.7B 8-bit fits with headroom; Voxtral 4-bit plus acoustic BF16 is a 16 GB machine’s upper TTS-only load if an LLM is also resident. iPhone 16 Pro: Kokoro CoreML at 0.08 RTF and 676 MB peak is the practical on-device narrator; Qwen3 CoreML is optional and heavier. Always leave vocoder and tail in higher precision when you quantize the talker.

The intern MP4 is fully synthetic stills held for seconds each. A spatial detector will likely flag faces as generated. A C2PA validator will find no manifest. Both results are consistent with the label on the set. They are not evidence of a leak and not evidence of an official briefing. The six source PDFs are not synthetic; running a deepfake detector on jp_hw.pdf is a category error. Secret-scanning the Pages HTML is reasonable. Do not paste API keys into letters.html.

The Japan letter is a self-classification against listed thresholds. It is not a license and not a commodity-jurisdiction determination for a particular deal, and the letter says so. Publishing it is republishing Apple’s own legal-resources PDF. The FY25 pCbCR Ireland line is Apple’s published table, with Apple’s own footnote on State Aid escrow. Neither document becomes an accusation by sitting next to OFAC’s framework. Adding a fabricated CEO to “support” the index would change republication into implied endorsement. That is another reason the video request was narrowed to a fictional intern.

23. What a future agent must not invent

Do not invent a sent email. Do not invent a gist URL. Do not invent a CIO workspace. Do not invent a Redis host. Do not invent CoreML benchmark numbers beyond the published harnesses cited here. Do not invent that an Apple chief executive appeared. Do not invent that isolation is still in artifacts/. Do not invent that C2PA signed the intern file. Do not invent that detectors cleared or failed a file that was not scored. The daily brief prompt already says this. The cost of a hallucinated send is higher than the cost of a repeated “still not sent.”

Compressed timeline. Morning into afternoon: ingest six PDFs; verify URLs; Pages index; letters; gist 403; Gmail timeout; user asks if mail went; answer no; research ESP and CDP chain through mParticle. Mid-afternoon: identity stitching, shared-device policy, session-scoped linking implemented and tested; CIO clocks documented; server isolation; Redis store, cache, eviction. Evening: operations log and PDF brief; daily automation; user deletes isolation code; weekly cleanup automation; Pages push of index and letters.html; CEO-anchored 1080p brief refused; intern desk accepted; stills and 25 s MP4; TTS landscape; quantization; MLX; CoreML; this deep dive. Cleanup of isolation and publication of Pages are the two durable mutations besides automations. The rest is record and refusal.

24. Quantization and ANE engineering notes

Group-wise affine quantization stores a scale (and optionally a zero-point) per group of weights, typically 64 elements in MLX dumps. The runtime dequantizes into the compute dtype before GEMM. That is why INT4 files are not automatically faster: the extra dequant can exceed the saved memory bandwidth on CPU, and on ANE the compiler may reject a graph that mixes palettes badly. Palettization (CoreML k-means) is different: weights become indices into a small codebook. It is closer

to a lookup table than to linear INT8. The Kokoro E2E INT8 file is palettized; the ONNX model_quantized.onnx is linear INT8. They are not drop-in replacements for each other and they do not share a runtime.

Fixed-shape CoreML graphs are why Kokoro E2E is capped at 5 s / 128 phonemes / 200 frames at 40 fps. The converter picked a shape the ANE compiler would accept. Longer audio is a Swift loop over that graph plus a crossfade, not a bigger tensor. Staged pipelines use buckets (T=32...512 for duration, 3 s...30 s for decoder) for the same reason. A “one model for all lengths” wish is how you get ANECCompile FAILED on iPhone while the Mac still runs.

computeUnits values: cpuOnly (simulator, debug), cpuAndGPU (escape hatch when ANE rejects), cpuAndNeuralEngine (the pin authors recommend first), all (scheduler may spill). Forcing all because “ANE is the point” is how you discover the scheduler already put your vocoder on GPU and your benchmark mixed two processors. Pin, measure per stage, then loosen.

G2P on device without espeak: dictionary (us_gold / us_silver), stemming, then a tiny BART-style CoreML encoder/decoder for OOV. That split exists so the iOS app is Apache-clean and does not ship GPL phonemizers. If you rebuild isolation-era session code, keep that same license hygiene. Do not pull Coqui XTTS into an iOS app.

MLX convert flags worth recording: --quantize, --q-bits 4|6|8, --q-group-size, --q-mode affine|mxfp4|mxfp8|nvfp4, --dtype float16|bfloat16|float32, --upload-repo. mxfp4 is a microscaling format, not “INT4 with extra steps.” If a dump sounds worse than affine 8-bit at the same size, try the other mode before blaming the model. Voxtral 4-bit explicitly leaves acoustic + vocoder in BF16; copying that pattern to CosyVoice (LLM+DiT INT4, HiFi-GAN fp32 with weight-norm folded) is the current good taste.

FAViT-style detectors that ingest an 11-channel forensic tensor (per-channel FFT magnitude, DWT sub-bands, noise residuals, Sobel, ELA) plus a ViT RGB branch are the 2026 academic shape. They still drop tens of points on DF26. Use them as a second opinion on a file you already distrust. Do not use them as a printer of certificates for the intern MP4.

25. Packet document close-reading — what the six files actually bind

Apple Japan letter (6 Mar 2026). A negative list: products not specified for use above 85 C or below -45 C; not radiation-hardened; not a machine above the stated WT class; not an inter-computer accessory above 2.0 GB/s in the listed sense; not systolic, neural, or optical computers in the control-list sense; not intrusion-program tooling; cryptography is mass-market, not user-modifiable, no vendor help required to use the crypto; not a quantum computer or part; theoretical peak performance below the stated 6,000 threshold. The letter covers “all Apple products” in that self-classification and tells the reader it is not a deal-by-deal ruling. Effect of quoting it as “Apple admits X is uncontrolled for this shipment to Y” would be a misread. Solution: quote the negatives and the disclaimer together.

EU pCbCR FY25. Entity-level public country-by-country figures in USD, not a 10-K consolidation. Ireland is the line that draws readers: revenue 213.56 billion, profit before tax 34.64 billion, cash tax 17.08 billion, current-year accrued 4.78 billion, 5,575 employees. Apple’s own narrative: cash tax ran above accrued mainly because State Aid escrow was released to Ireland, with the expense booked in FY24. Effect of quoting cash tax without the escrow sentence is a distortion. Solution: keep the footnote welded to the number, as the live index already does.

OFAC Framework (2020). Five components of a sanctions compliance program: management commitment, risk assessment, internal controls, testing and auditing, training. It is Treasury guidance on how to build a program, not a finding that any company in the packet failed one. Binding it to Apple Japan’s letter in one index is juxtaposition, not an OFAC action.

CISTEC comments (17 Jan 2023) sit on docket BIS-2022-0029, which is closed. They cannot be refiled as a new comment through the index. Republishing a closed-docket PDF is documentation, not participation in the rulemaking.

Yuan and Baspin, arXiv 2605.18961. Explicit constructions of 4D and 5D CSS Layer Codes by color routing. With good qLDPC input they saturate the corresponding BPT bounds. This is an open-access physics/CS paper. It is in the packet because the operator put it there, not because Apple’s Japan letter is a quantum-computer classification that the paper contradicts. The letter already says Apple products are not quantum computers. The paper is about codes, not Apple SKUs. Keep that distinction on the index.

Alabama affidavit. Beason-Hammon / Ala. Code 31-13-9 subcontractor immigration-compliance form, blank. Not executed. Not Apple. Not federal. It is the one item whose presence in an “Apple, politics, and the public” bundle most needs a one-line justification on the page. Until that line exists, split it out of the send text.

Together the six files are a reading list. They become a claim only if a human writes a claim on top. The intern video, if voiced, should read the index’s banner (“Not a leak”) and the URLs, not a new theory of why the six belong together. The deep dive you are reading is allowed to say “juxtaposition risk” because that is an operational finding, not a hidden plot.

26. Appendix — commands, prompts, and copy-ready send text

26.1 Live URLs

Public index: <https://superrep2.github.io/packet-2026-09-24.html>. Copy-ready letters: <https://superrep2.github.io/letters.html>. Site home: <https://superrep2.github.io/>. Control-plane repo (secondary copy): <https://github.com/superrep2/qsrc-control-plane>. Pages repo: <https://github.com/superrep2/superrep2.github.io>. Operator: <https://x.com/DimensionsAlgo>. Apple Japan source: https://www.apple.com/legal/more-resources/docs/jp_hw.pdf. pCbCR hub: <https://www.apple.com/legal/more-resources/>. OFAC: <https://ofac.treasury.gov/media/30776/download?inline>. CISTEC docket: <https://www.regulations.gov/docket/BIS-2022-0029>. Paper: <https://arxiv.org/abs/2605.18961>.

26.2 ffmpeg used for the intern package

Stills scaled with `scale=1920:1080:force_original_aspect_ratio=decrease, pad=1920:1080:(ow-iw)/2:(oh-ih)/2, setsar=1, format=yuv420p`. Concat: title 4 s, intern 8 s, wide desk 8 s, end card 5 s. Silent bed: `anullsrc r=48000 stereo`. Video fade in 0.4 s, fade out last 0.5 s. Encoder: `libx264 crf 20 preset medium threads 4, aac 96k, pix_fmt yuv420p, movflags +faststart`. Output 1920x1080, 30 fps, 25.00 s, ~333 kb/s video. Path: `artifacts/packet-brief-fictional-intern-1080p.mp4`.

26.3 Mux recipe when a catalog voice exists

`ffmpeg -n -i artifacts/packet-brief-fictional-intern-1080p.mp4 -i artifacts/intern-voice.mp3 -filter_complex "[1:a]afade=t=in:st=0:d=0.2,afade=t=out:st=24.3:d=0.5[a]" -map 0:v -map "[a]" -c:v copy -c:a aac -b:a 128k -shortest -movflags +faststart artifacts/packet-brief-fictional-intern-voiced-1080p.mp4`. First spoken line must disclose the fictional desk. Do not time-stretch the stills to match a three-minute script without generating more cards; the current picture pack is 25 seconds.

26.4 Weekly cleanup prompt (verbatim substance)

Weekly isolation-code cleanup. Delete only the session-isolation package if it exists. Remove `artifacts/identity-session-link/` and any recreation of `session-link.js`, `server-isolation.js`, `redis-cache.js`, `redis-store.js`, `rudderstack-transform.js`, `segment-insert.js`, `warehouse-session-link.sql`, `client-session.js`, `cio-pipelines.js` under that folder. Do not delete packet letters, operations logs, or PDF briefs. Do not touch automations, bots, GitHub, or production CIO/Segment/RudderStack. If the folder is already gone, say so and stop.

26.5 Daily brief prompt (verbatim substance)

Generate today's operations log and a one-to-two-page PDF brief. Date files YYYY-MM-DD in America/Chicago under artifacts/. Cover what actually happened since the previous brief: packet/outreach status, connectors, CDP/identity/CIO/Redis work, code under artifacts/identity-session-link/, tests, open items. Do not invent sends or deployments. Rank every item 0–5 (5 = wrong-person messaging, consent break, or irreversible identity merge). Restrained editorial, no purple or neon. Carry forward unresolved items from artifacts/2026-09-24-operations-log.md until closed.

26.6 Copy-ready Apple media body

To: media.help@apple.com. Subject: Public source packet — Japan classification, EU pCbCR FY25, OFAC, Layer Codes. Body: Public index, not a leak: <https://superrep2.github.io/packet-2026-09-24.html>. Apple Japan 6 Mar 2026 letter: https://www.apple.com/legal/more-resources/docs/jp_hw.pdf. Apple EU pCbCR hub: <https://www.apple.com/legal/more-resources/>. OFAC SCP framework: <https://ofac.treasury.gov/media/30776/download?inline>. CISTEC / BIS-2022-0029 (closed docket): <https://www.regulations.gov/docket/BIS-2022-0029>. Yuan and Baspin arXiv:2605.18961: <https://arxiv.org/abs/2605.18961>. Please confirm the Japan letter and FY25 pCbCR are still current.

26.7 Copy-ready X draft

Public source packet, 24 Sep 2026 — not a leak. Apple Japan hardware classification letter, EU pCbCR FY25, OFAC SCP framework, CISTEC export-control comments, Iceberg Quantum Layer Codes. Index: <https://superrep2.github.io/packet-2026-09-24.html>

26.8 Physical lines

Apple Inc., One Apple Park Way, Cupertino, CA 95014. Apple Japan, Inc., 6-10-1 Roppongi, Minato-ku, Tokyo 106-6140. House finder: <https://www.house.gov/representatives/find-your-representative>. Senate contacts: <https://www.senate.gov/senators/senators-contact.htm>.

26.9 Deleted isolation module inventory (so it is not mysterious later)

session-link.js (planIdentify, ensureFreshSession, eventInSessionScope). server-isolation.js (createIsolationServer, buildTrack, buildIdentify, isSpent, retireAnon). client-session.js (kiosk skips sessionStorage). cio-pipelines.js. segment-insert.js. rudderstack-transform.js. warehouse-session-link.sql (login graph only). redis-store.js (TTL adapter). redis-cache.js (L1 LRU + L2 cache-aside). redis-example.js. Tests: session-link, server-isolation, redis-store, redis-cache. README, package.json. All removed from artifacts/identity-session-link/ on user instruction. Weekly job will remove them again if they return to that path.

26.10 Connected tools that existed in this chat

GitHub (push, get contents, tree, search, get_me; gist 403). Voice (generate_speech, list_voices, generate_multi_speech). Automations (list, create, update, pause, validate). Not connected for mail: Gmail timed out; Outlook unused. Not connected: SendGrid, Customer.io, Segment, RudderStack, Redis Cloud. Linux sandbox tools used: reportlab, ffmpeg, pypdf.

27. Closing judgment

The packet is public. The mail is not sent. The isolation code is gone by request and scheduled to stay gone. The dangerous clocks at Customer.io are unchanged because they never lived in this folder. The dangerous video was not made. The safe video was made and labeled. Speech research says a catalog voice on the Voice connector, or Kokoro on a Mac ANE, is enough to finish the intern package without impersonation. Quantization research says INT8 and a float vocoder; INT4 is

often a vanity file. Provenance research says C2PA will not save a cropped deepfake and is not required to explain a labeled illustration.

The session's coherent policy, stated without marketing language: publish official sources with URLs; do not pretend a Pages site is a summons; do not merge strangers on shared glass; do not mint official-looking media of living executives; when you illustrate, write FICTION on the set; when you speak, use a voice that belongs to no one in particular; when you delete a safeguard, remember the vendor still has the merge.

Next brief (daily automation) should record this PDF's path, the intern MP4, the weekly cleanup id, and the still-open mail item at severity 3. It should not record a send that did not happen and should not restore identity-session-link unless the user pauses the Thursday job and asks for the code back.

End of working record. Assembled 24 September 2026. Not a leak. Not a broadcast. Not legal advice.